10. In frequency modulation (FM), the modulated signal $y(t)$ is related to the modulating signal $m(t)$ by

$$y(t) = A\cos\left(\omega_c t + \omega_\Delta \int_{-\infty}^{t} m(\tau)d\tau\right)$$

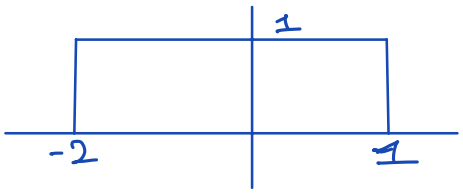
where ω_Δ is the frequency-deviation constant. This is called FM because the instantaneous frequency is proportional to the modulating signal:

$$\omega(t) = \frac{d}{dt}\left[\omega_c t + \omega_\Delta \int_{-\infty}^{t} m(\tau)d\tau\right]$$
$$= \omega_c + \omega_\Delta m(t).$$

(a) Sketch $y(t)$ for $\omega_c = 8\pi$, $\omega_\Delta = 2\pi$ and $m(t) = u(t+2) - u(t-1)$.

(b) Is the modulation system, with input $m(t)$ and output $y(t)$, linear? Time invariant? Memoryless? Causal?

$m(t)=$



$y(t) = A\cos\left(8\pi t + 2\pi \int_{-\infty}^{t} m(t)\right)$

(i) $t \le -2$

$y(t) = A\cos(8\pi t)$

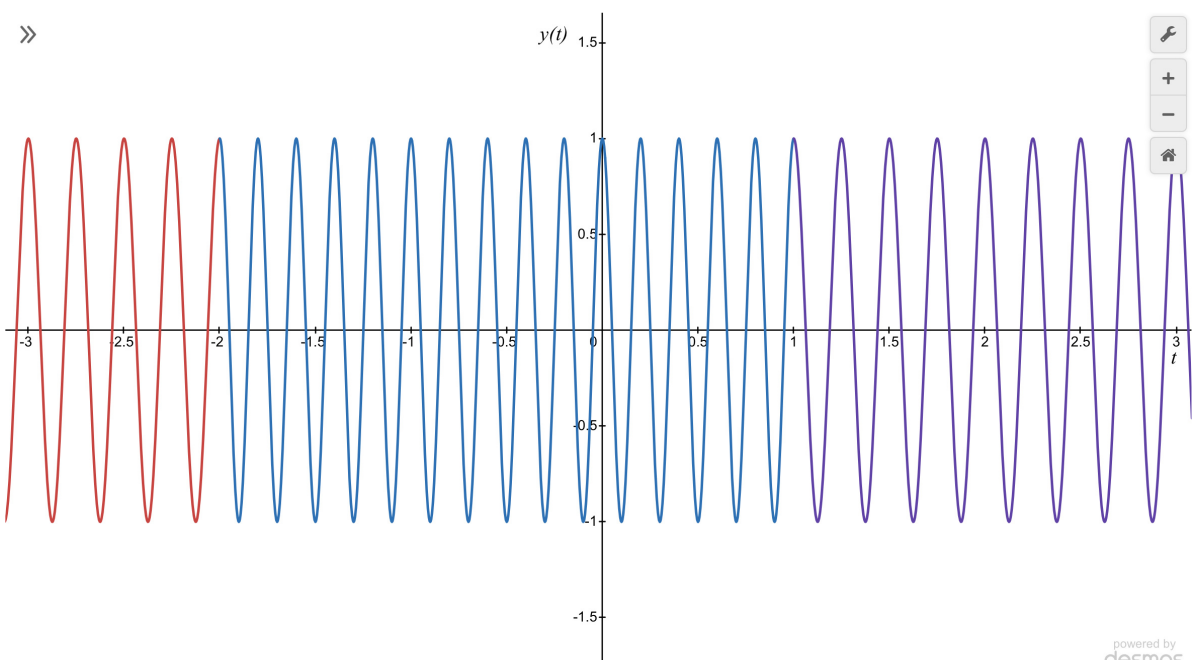
(ii) $t \in (-2, 1)$

$y(t) = A\cos[10\pi t + 4\pi]$

$= A\cos(10\pi t)$

(iii) $t \ge 1$

$y(t) = A\cos(8\pi t)$



(b) The system is non-linear (cosine), time variant (phase shift in cosine), has memory (depends on past), causal (doesn't depend on future)